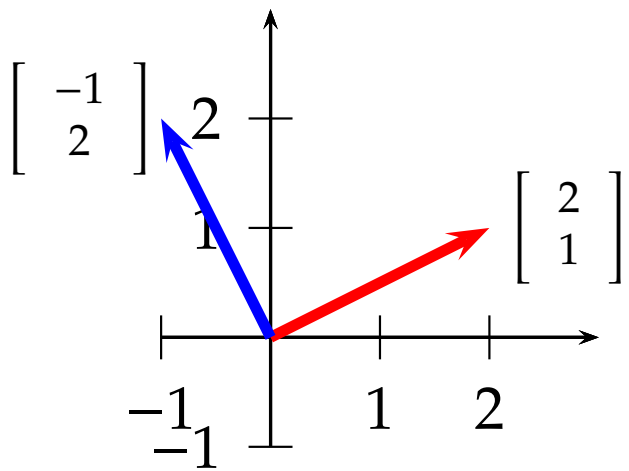
Vectors

An $n \times 1$ matrix is called a vector or a column vector.

Vectors can be realized geometrically :

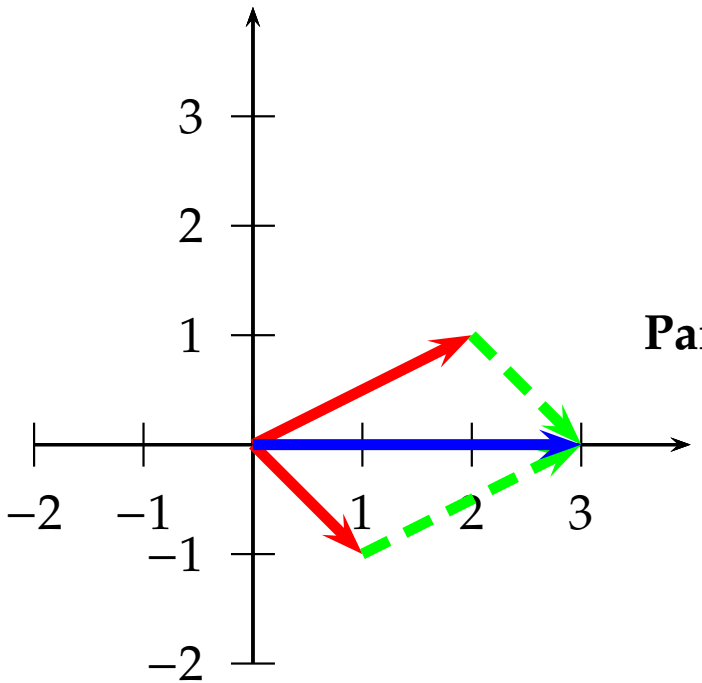


We can identify $\mathbb{R}^n$ with the set of all $n \times 1$ vectors.

Vector addition

Given two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ (or $\mathbb{R}^n$), we can take their sum:

$$\mathbf{u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \implies \mathbf{u} + \mathbf{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$



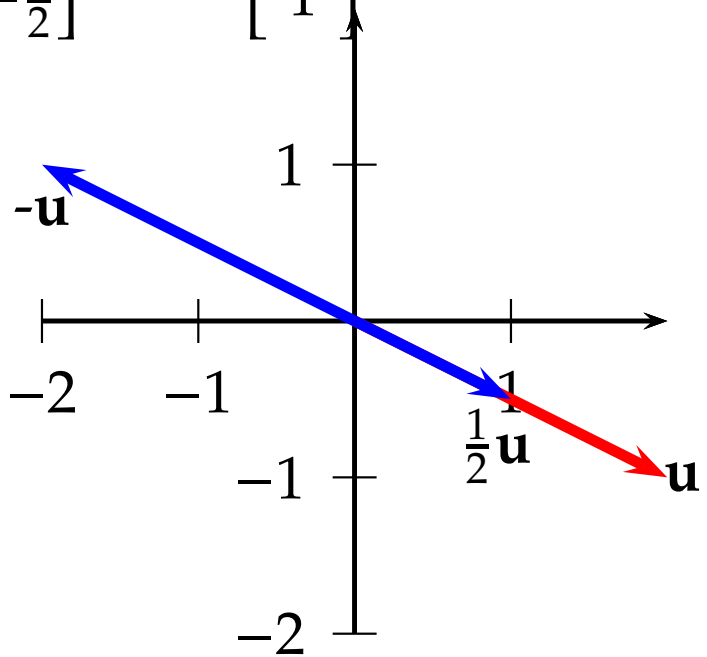
Parallelogram Law

Scalar multiplication

We can also rescale vectors by any scalar.

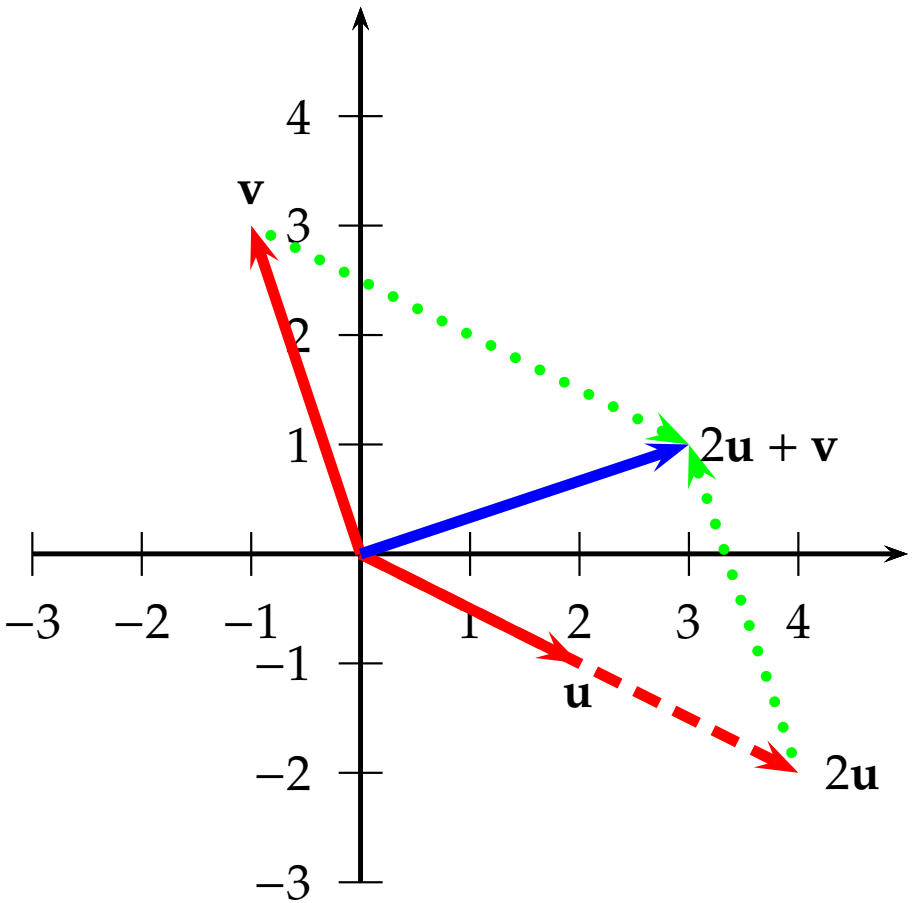
$$2\begin{bmatrix} 1 \\ 3 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 0 \\ -2 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \tfrac{1}{2}\mathbf{u} = \begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix} \quad -\mathbf{u} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$



$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$2\mathbf{u} + \mathbf{v} = 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$



Elementary properties of vector operations

Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are vectors in $\mathbb{R}^n$, and c, d are scalars. Then :

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
4. $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$
5. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
6. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
7. $c(d\mathbf{u}) = (cd)\mathbf{u}$
8. $1\mathbf{u} = \mathbf{u}$

Linear Combinations

Given vectors $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_p$ in $\mathbb{R}^n$, any vector

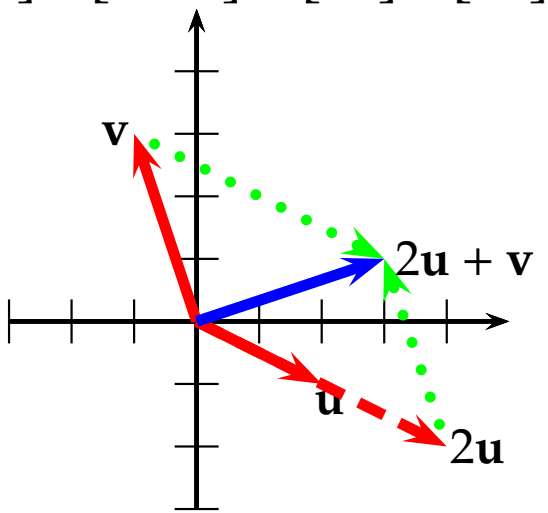
$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

where $c_1, ..., c_p$ are scalars is called a linear combination of $\mathbf{v}_1, ..., \mathbf{v}_p$ with weights $c_1, ..., c_p$.

Example.

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$2\mathbf{u} + \mathbf{v} = 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$



Example

Let

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ -1 \\ -2 \\ 1 \end{bmatrix}$$

and $c_1 = -1, c_2 = 2, c_3 = -2$.

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = (-1)\begin{bmatrix} 3 \\ -1 \\ 2 \\ -3 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} + (-2)\begin{bmatrix} 0 \\ -1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \\ 5 \end{bmatrix}$$

$\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ with weights $c_1 = -1, c_2 = 2, c_3 = -2$.

If $\mathbf{v}_1, ..., \mathbf{v}_p$ are vectors in $\mathbb{R}^n$, then any linear combination of $\mathbf{v}_1, ..., \mathbf{v}_p$ is also a vector in $\mathbb{R}^n$.

Different linear combinations of the same list of vectors can be equal to the same vector:

Example

$$\begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - 1\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

• **Question.** Given $\mathbf{b}$ and $\mathbf{a}_1, \ldots, \mathbf{a}_p$ in $\mathbb{R}^n$, how can we determine if $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_p$?

Example

Determine if $\mathbf{b} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix}$ is a linear combination of $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$.

Solution. We need to find scalars x_1, x_2 such that

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b}.$$

In other words we need to solve the above "vector equation" for x_1, x_2.

$$x_1\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + x_2\begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix} \implies \begin{bmatrix} x_1 \\ -x_1 \\ 3x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 2x_2 \\ -x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix}$$

$$\implies \begin{bmatrix} x_1 + 2x_2 \\ -x_1 + 2x_2 \\ 3x_1 - x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix} \implies \begin{cases} x_1 & +2x_2 & = 0 \\ -x_1 & +2x_2 & = -4 \\ 3x_1 & -x_2 & = 7 \end{cases}$$

Example

Determine if $\mathbf{b} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix}$ is a linear combination of $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$.

We should solve the linear system:

$$\begin{cases} x_1 & +2x_2 & = 0 \\ -x_1 & +2x_2 & = -4 \\ 3x_1 & -x_2 & = 7 \end{cases} \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 2 & -4 \\ 3 & -1 & 7 \end{array}\right] \text{ augmented matrix}$$

$$\left[\begin{array}{cc|c} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{b} \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 2 & -4 \\ 3 & -1 & 7 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 4 & -4 \\ 3 & -1 & 7 \end{array}\right] \xrightarrow{R_3 \to R_3 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 4 & -4 \\ 0 & -7 & 7 \end{array}\right] \xrightarrow{R_2 \to \frac{1}{4}R_2}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -7 & 7 \end{array}\right] \xrightarrow{R_3 \to R_3 + 7R_2} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \text{ E.F. } \xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \text{ R.E.F.}$$

Example

Determine if $\mathbf{b} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix}$ is a linear combination of $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$.

We we did so far:

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 = \mathbf{b} \rightsquigarrow \begin{cases} x_1 & +2x_2 & = 0 \\ -x_1 & +2x_2 & = -4 \\ 3x_1 & -x_2 & = 7 \end{cases} \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 2 & -4 \\ 3 & -1 & 7 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 2 & -4 \\ 3 & -1 & 7 \end{array}\right] \xrightarrow{\text{Row reduction}} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \text{ R.E.F. } \rightsquigarrow \begin{cases} x_1 & & = 2 \\ & x_2 & = -1 \end{cases}$$

Therefore we have

$$2\mathbf{a}_1 + (-1)\mathbf{a}_2 = \mathbf{b}$$

or

$$2\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix} + (-1)\begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \\ 7 \end{bmatrix}$$

Let $\mathbf{a}_1, \ldots, \mathbf{a}_p$ and $\mathbf{b}$ be vectors in $\mathbb{R}^n$.

- The vector equation

$$x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right]$$

- In particular, $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \ldots, \mathbf{a}_p$ if and only if the linear system with the above augmented matrix is consistent.

Example

Determine if $\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ is a linear combination of $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}$.

$$x_1 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} x_1 - 2x_2 + 3x_3 \\ -2x_1 + x_2 - 3x_3 \\ x_1 + x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ -2 & 1 & -3 & 3 \\ 1 & 1 & 0 & -1 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \text{R.E.F.}$$

The system is inconsistent.

$\begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ is <u>not</u> a linear combination of $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 3 \\ -3 \\ 0 \end{bmatrix}$.

Example

Determine if $\begin{bmatrix}-2\\3\\-1\end{bmatrix}$ is a linear combination of $\begin{bmatrix}1\\-2\\1\end{bmatrix}$, $\begin{bmatrix}-2\\1\\1\end{bmatrix}$, and $\begin{bmatrix}3\\-3\\0\end{bmatrix}$.

$$x_1\begin{bmatrix}1\\-2\\1\end{bmatrix}+x_2\begin{bmatrix}-2\\1\\1\end{bmatrix}+x_3\begin{bmatrix}3\\-3\\0\end{bmatrix}=\begin{bmatrix}-2\\3\\-1\end{bmatrix}\rightsquigarrow\begin{bmatrix}x_1-2x_2+3x_3\\-2x_1+x_2-3x_3\\x_1+x_2\end{bmatrix}=\begin{bmatrix}-2\\3\\-1\end{bmatrix}$$

$$\left[\begin{array}{ccc|c}1&-2&3&-2\\-2&1&-3&3\\1&1&0&-1\end{array}\right]\xrightarrow{\text{Row Reduction}}\left[\begin{array}{ccc|c}1&0&1&-\frac{4}{3}\\0&1&-1&\frac{1}{3}\\0&0&0&0\end{array}\right]\text{ R.E.F.}$$

$$\begin{array}{l}x_1,x_2:\text{basic}\\x_3:\text{free}\end{array}\rightsquigarrow\begin{cases}x_1+x_3=-\frac{4}{3}\\x_2-x_3=\frac{1}{3}\\0=0\end{cases}\Rightarrow\begin{cases}x_1=-x_3-\frac{4}{3}\\x_2=x_3+\frac{1}{3}\\x_3:\text{free}\end{cases}$$

Conclusion: $\begin{bmatrix}-2\\3\\-1\end{bmatrix}$ is a linear combination of $\begin{bmatrix}1\\-2\\1\end{bmatrix}$, $\begin{bmatrix}-2\\1\\1\end{bmatrix}$, and $\begin{bmatrix}3\\-3\\0\end{bmatrix}$.

Example. $x_3=1\Rightarrow x_1=-\frac{7}{3}$ and $x_2=\frac{4}{3}\rightsquigarrow(-\frac{7}{3})\begin{bmatrix}1\\-2\\1\end{bmatrix}+\frac{4}{3}\begin{bmatrix}-2\\1\\1\end{bmatrix}+1\begin{bmatrix}3\\-3\\0\end{bmatrix}=\begin{bmatrix}-2\\3\\-1\end{bmatrix}$.

Span$\{\mathbf{v}_1, ..., \mathbf{v}_p\}$

Suppose we are given vectors $\mathbf{v}_1, ..., \mathbf{v}_p$ in $\mathbb{R}^n$.

- The set of all linear combinations of $\mathbf{v}_1, ..., \mathbf{v}_p$ is called the **subset of $\mathbb{R}^n$ spanned (or generated) by $\mathbf{v}_1, ..., \mathbf{v}_p$.**

This subset is represented by

$$\text{Span}\{\mathbf{v}_1, ..., \mathbf{v}_p\}$$

- Therefore, Span$\{\mathbf{v}_1, ..., \mathbf{v}_p\}$ contains all vectors $\mathbf{w}$ of the form

$$\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p$$

where $c_1, ..., c_p$ are arbitrary scalars.

Example

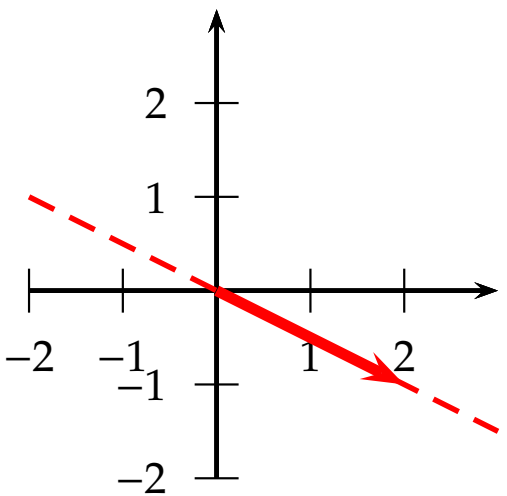
Let $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Describe Span$\{\mathbf{v}\}$ both algebraically and geometrically.

Solution. Algebraically, Span$\{\mathbf{v}\}$ is the set of all vectors of the form $c\mathbf{v}$. Note that:

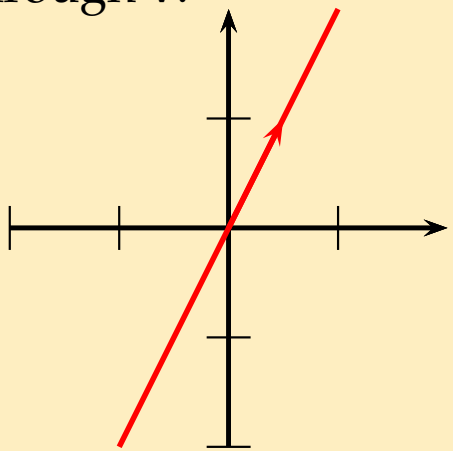
$$c\mathbf{v} = c\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2c \\ -c \end{bmatrix}$$

$$\text{Span}\{\mathbf{v}\} = \{c \cdot \mathbf{v} \,:\, c \in \mathbb{R}\} = \left\{\ldots, \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \ldots, \begin{bmatrix} 4 \\ -2 \end{bmatrix}, \ldots, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \ldots, \begin{bmatrix} -1 \\ \frac{1}{2} \end{bmatrix}, \ldots, \begin{bmatrix} -\sqrt{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \ldots\right\}$$

Geometrically, Span$\{\mathbf{v}\}$ is the set of all vectors that lie on the line through $\mathbf{v}$.



In general, if $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^2$ (or $\mathbb{R}^n$) then Span$\{\mathbf{v}\}$ is the set of all vectors that lie on the line through $\mathbf{v}$.

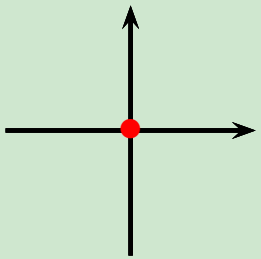


Example

What is Span$\{\mathbf{v}\}$ where $\mathbf{v} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$?

Solution. It is just the zero vector. Algebraically, it is $\left\{\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}$.

Geometrically, it is:

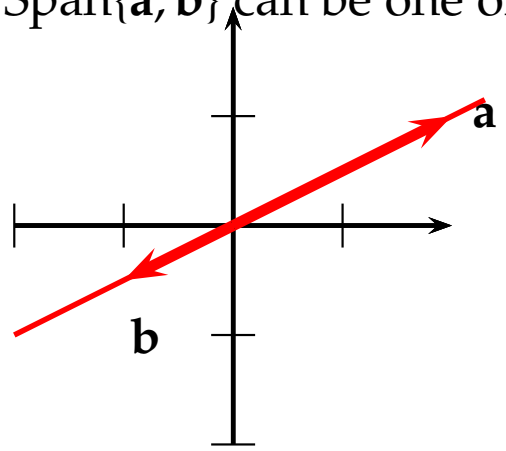


If $\mathbf{0}$ denotes the zero vector in $\mathbb{R}^n$, then Span$\{\mathbf{0}\} = \{\mathbf{0}\}$, i.e., just the origin.

Geometric description of Span{**a**, **b**}

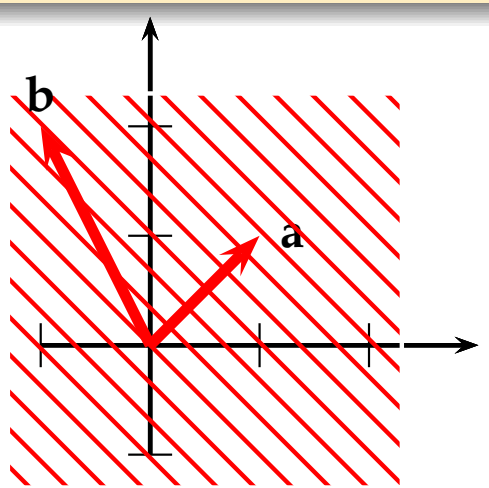
• If **a** and **b** are two nonzero vectors in $\mathbb{R}^2$, then Span{**a**, **b**} can be one of the following:

1. If **a** and **b** are parallel, then Span{**a**, **b**} is the line through **a** and **b**.



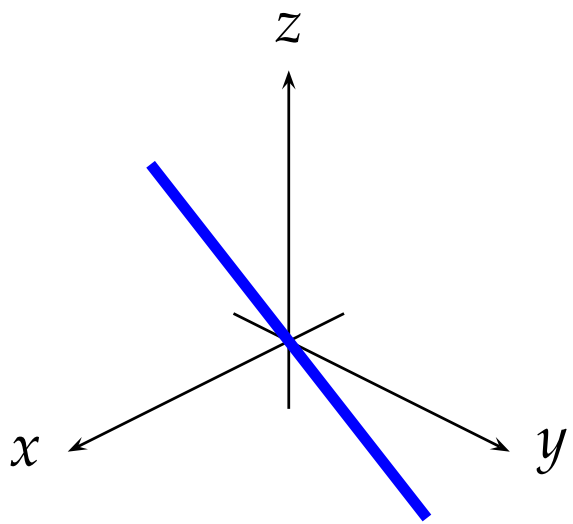
Two vectors **v**, **w** in $\mathbb{R}^n$ are called parallel if either **v** = c**w** or **w** = c**v** for some scalar c.

2. Otherwise, Span{**a**, **b**} is all of $\mathbb{R}^2$.

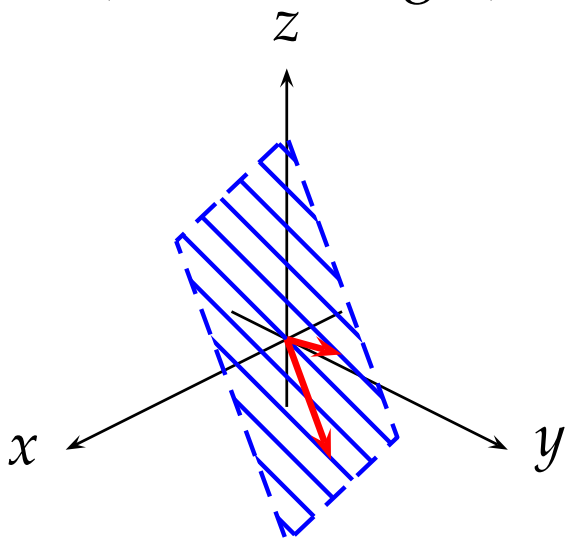


Similarly, for two nonzero vectors $\mathbf{a}, \mathbf{b}$ in $\mathbb{R}^3$, Span$\{\mathbf{a}, \mathbf{b}\}$ is one of the following.

1. If $\mathbf{a}$ and $\mathbf{b}$ are parallel, then Span$\{\mathbf{a}, \mathbf{b}\}$ is the line passing through $\mathbf{a}$ and $\mathbf{b}$ (and the origin).



2. If $\mathbf{a}$ and $\mathbf{b}$ are not parallel, then $\text{Span}\{\mathbf{a}, \mathbf{b}\}$ is the plane containing $\mathbf{a}$ and $\mathbf{b}$ (and the origin)



The punchline...

1. Given a linear system, we can write an equivalent vector equation, and vice versa.

   Example.

$$\begin{cases} -x_1 & & +2x_3 & -2x_4 & = 1 \\ & x_2 & +x_3 & -x_4 & = -1 \\ x_1 & +5x_2 & -4x_3 & +x_4 & = 3 \end{cases}$$

   with augmented matrix

$$\left[\begin{array}{cccc|c} -1 & 0 & 2 & -2 & 1 \\ 0 & 1 & 1 & -1 & -1 \\ 1 & 5 & -4 & 1 & 3 \end{array}\right]$$

   can be written as :

$$x_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 1 \\ -4 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$$

2. The linear system and the vector equation have the *same* solution set.

Given vectors $\mathbf{b}$ and $\mathbf{a}_1, \ldots, \mathbf{a}_p$ in $\mathbb{R}^n$, the following statements are equivalent (i.e., if any of them is true, then the rest are true):

- $\mathbf{b}$ is a linear combination of $\mathbf{a_1}, ..., \mathbf{a_p}$.
- $\mathbf{b} = c_1\mathbf{a}_1 + \cdots + c_p\mathbf{a}_p$ for some scalars $c_1, ..., c_p$.
- The vector equation $x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$ has a solution.
- the linear system with augmented matrix

$$\left[\begin{array}{ccc|c} \mathbf{a}_1 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right]$$

  is consistent.